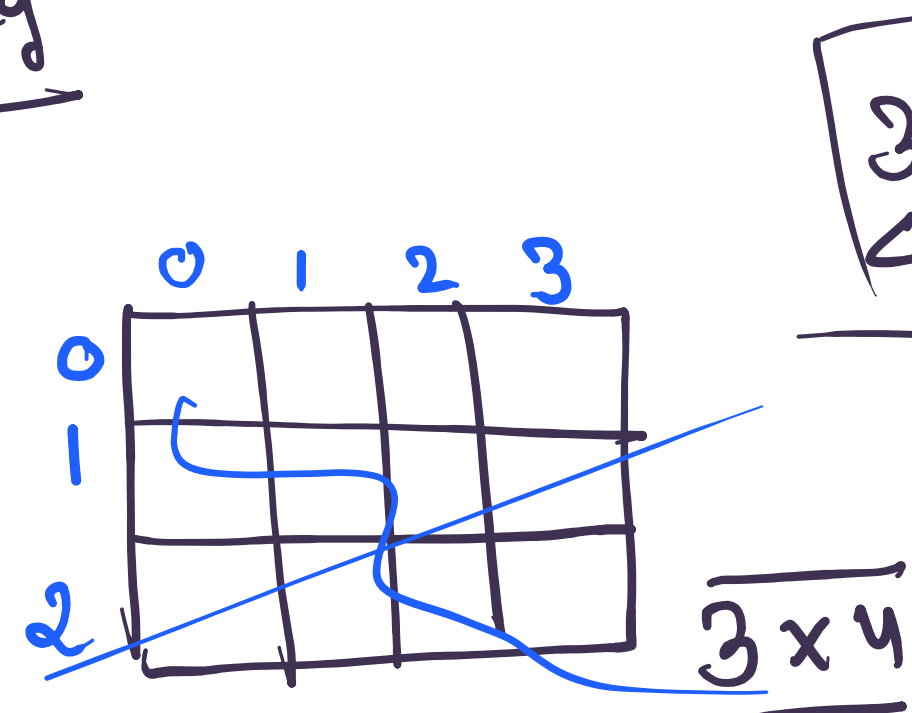
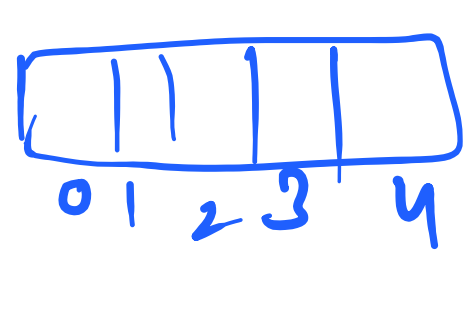


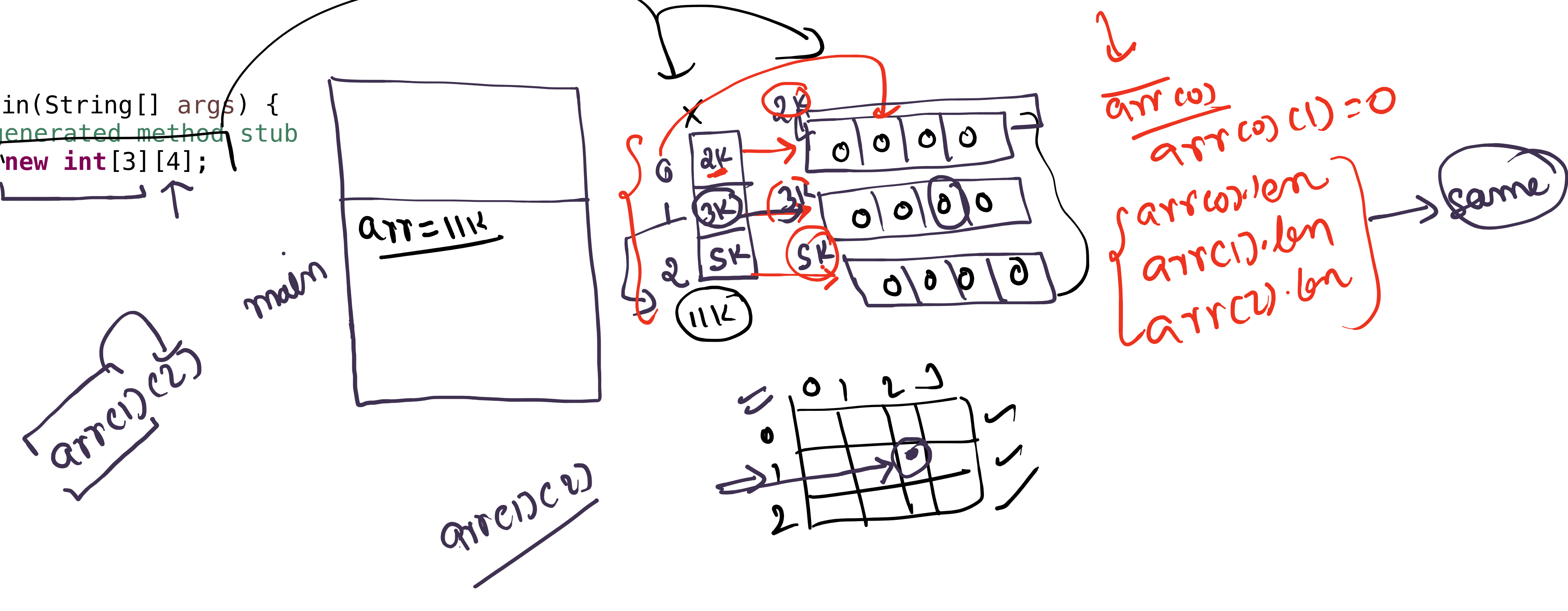
2D-Array



3 → Row
4 → col

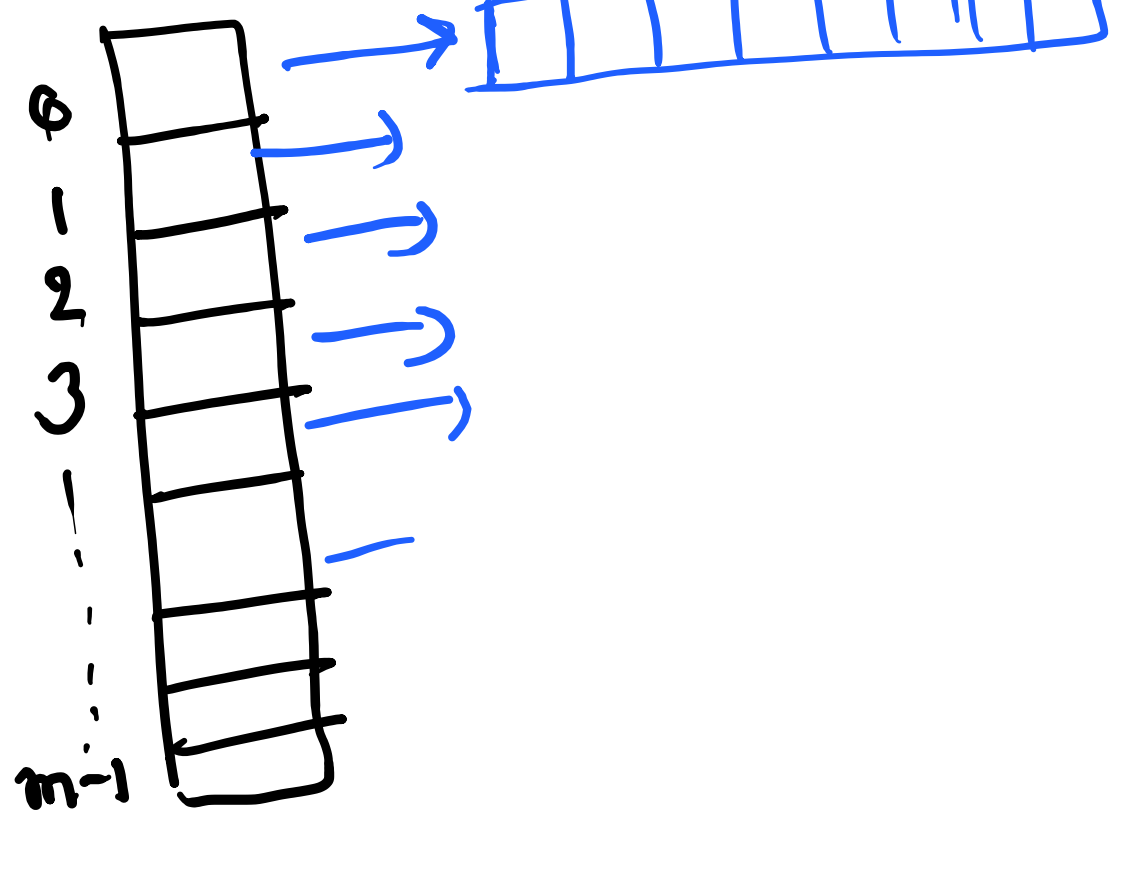


```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int[][] arr = new int[3][4];  
}
```



```
int C[] arr = new int [M][N]
```

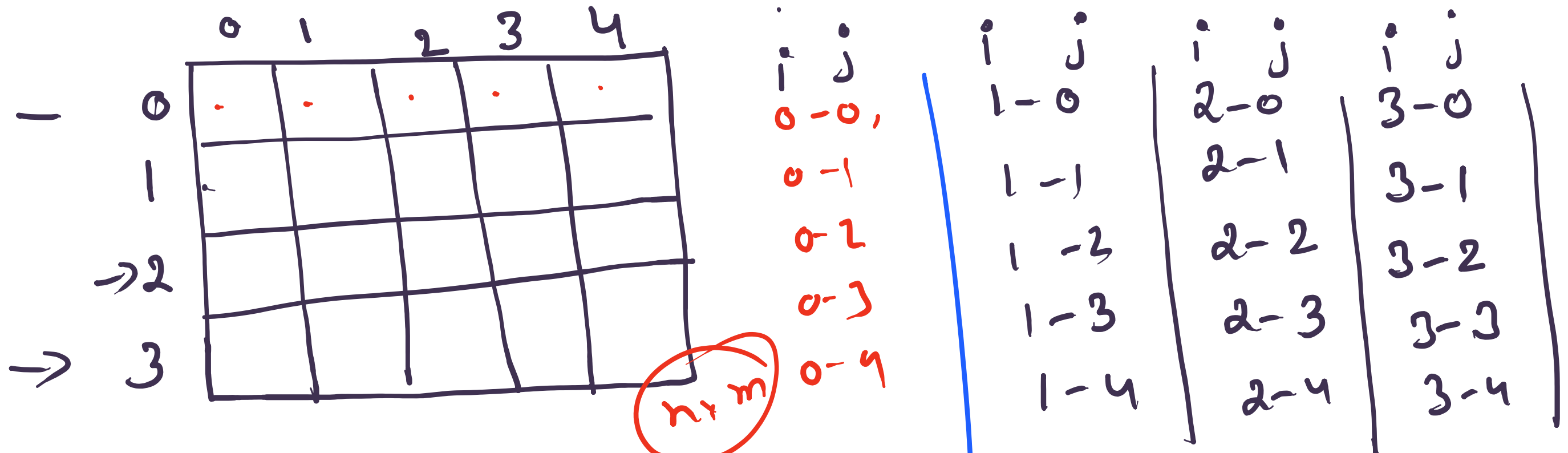
1 + m



for (i=0; i<arr.len; i++)

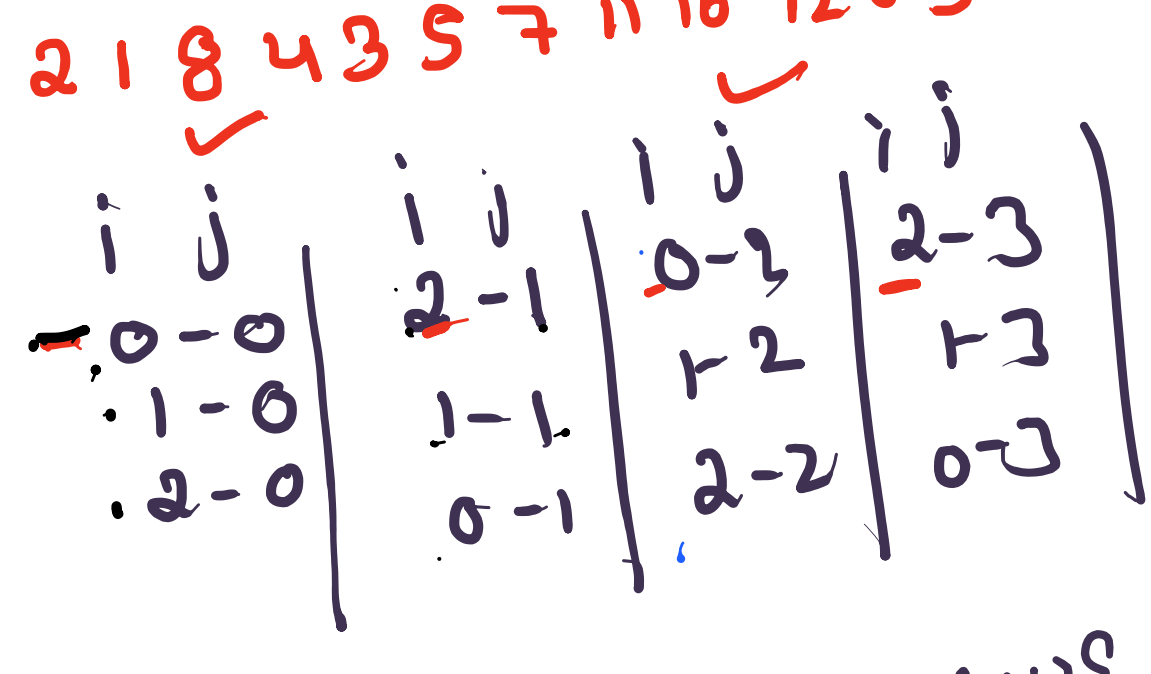
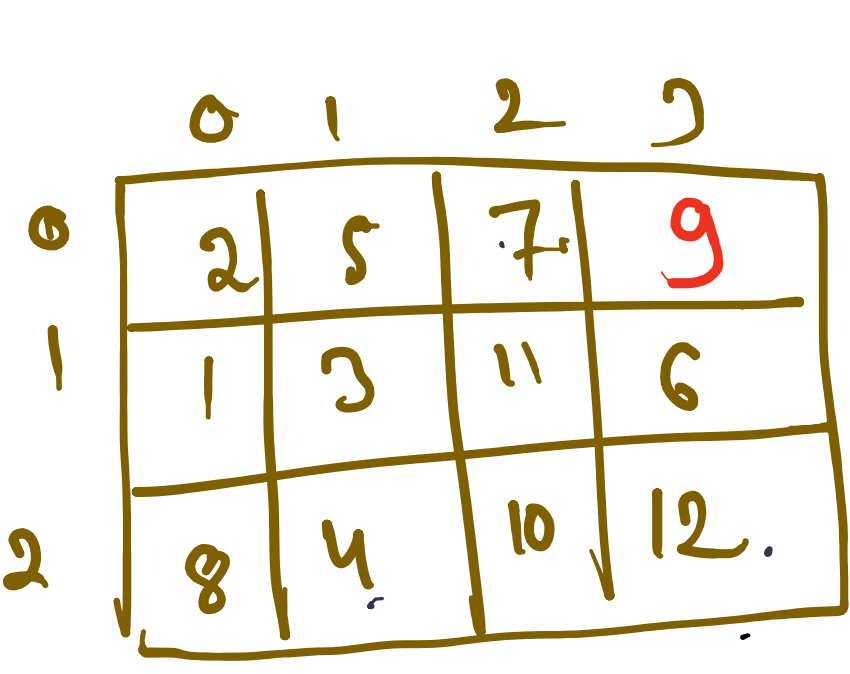
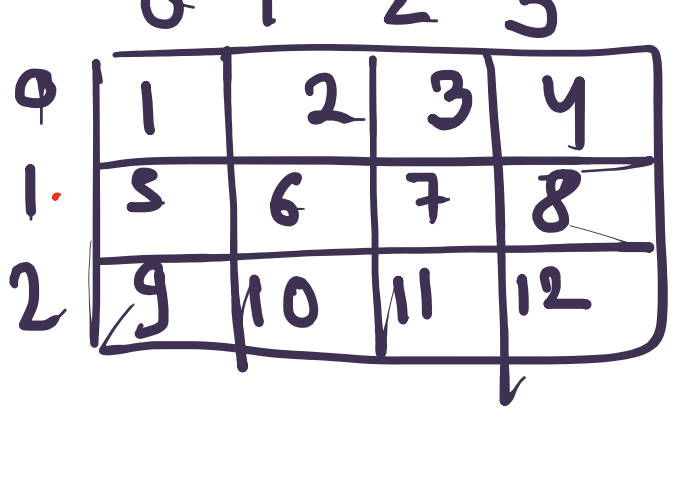
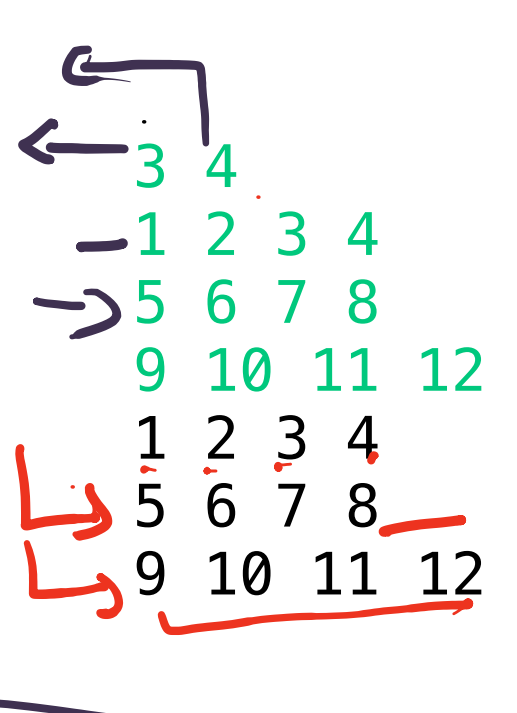
arr.length

Row = 2
Col = 1



for (i=0; i<arr.len; i++)
for (j=0; j<arr.col.len; j++)
arr[i][j] = sc.nextInt();

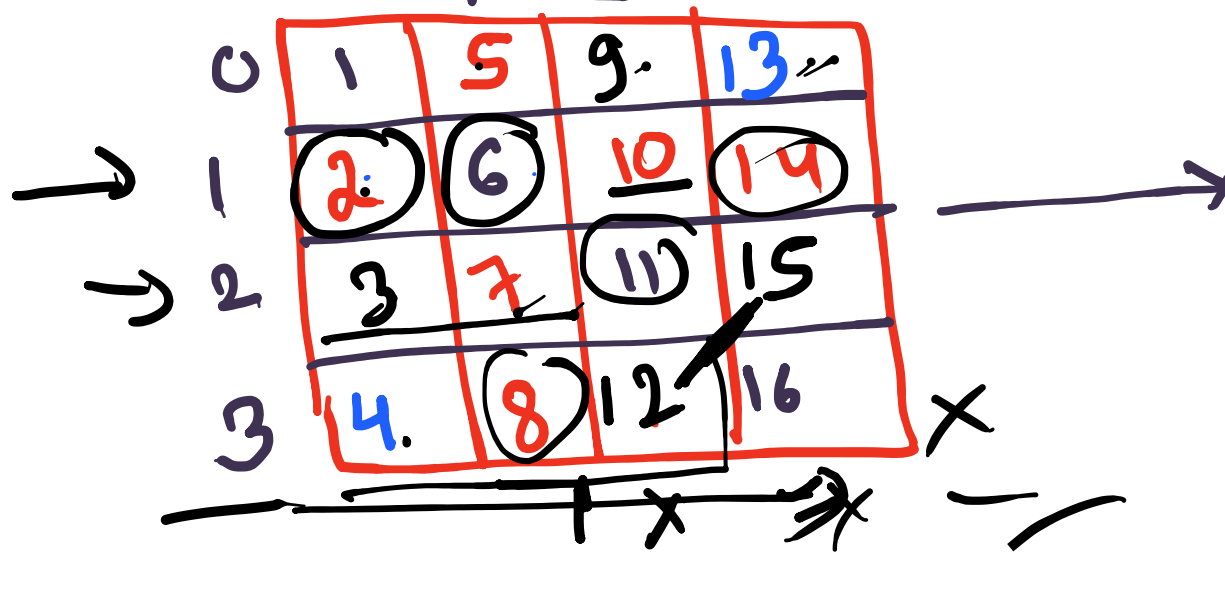
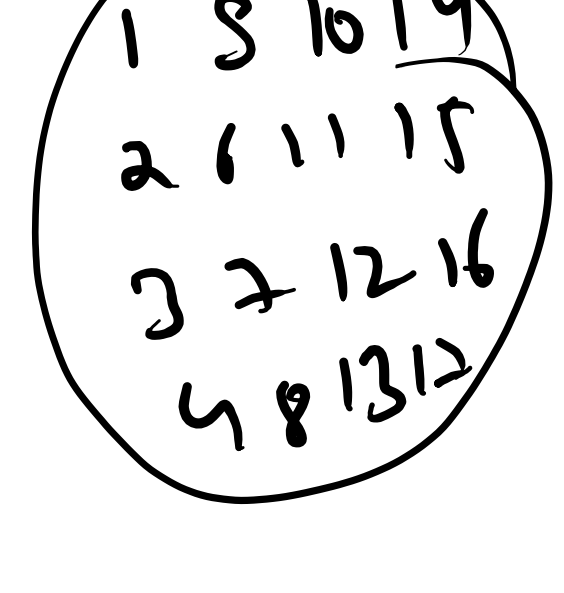
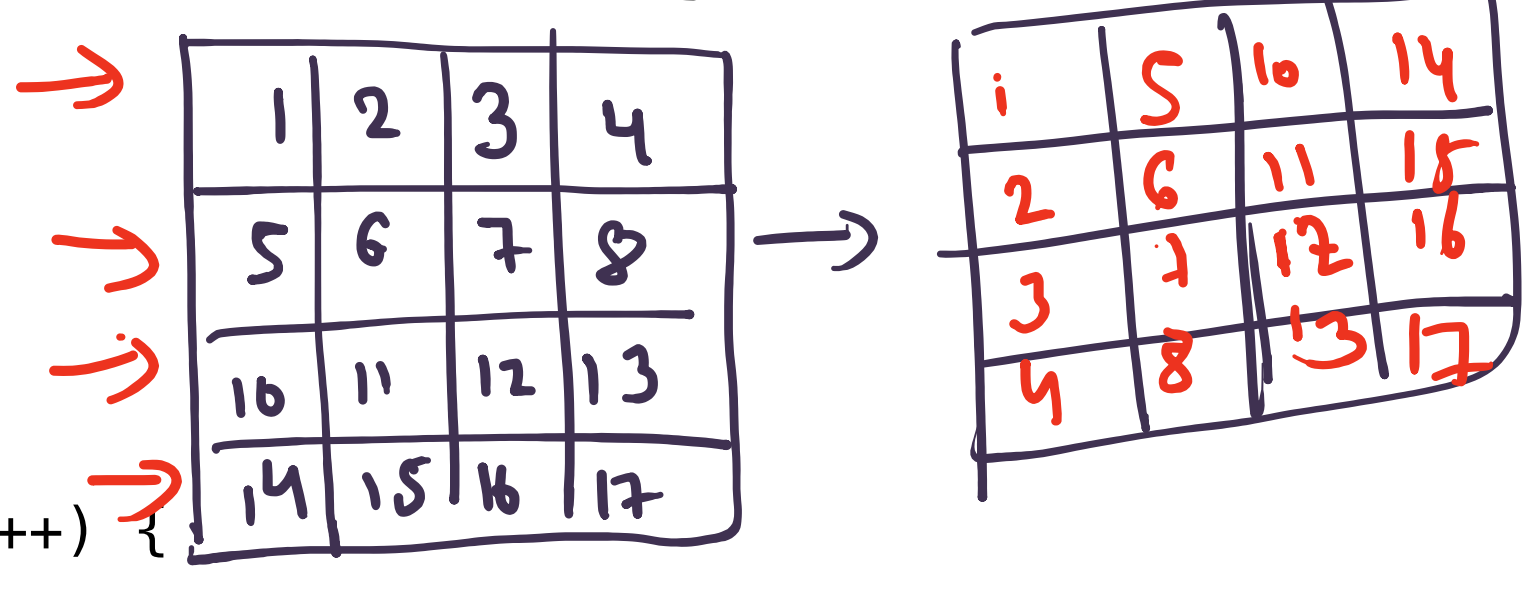
```
import java.util.Scanner;  
public class Input_Output {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int m = sc.nextInt();  
        int[][] arr = new int[n][m];  
        for (int i = 0; i < arr.length; i++) {  
            for (int j = 0; j < arr[0].length; j++) {  
                arr[i][j] = sc.nextInt();  
            }  
        }  
        Display(arr);  
    }  
    public static void Display(int[][] arr) {  
        for (int i = 0; i < arr.length; i++) {  
            for (int j = 0; j < arr[0].length; j++) {  
                System.out.print(arr[i][j] + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```



arr[i][j]

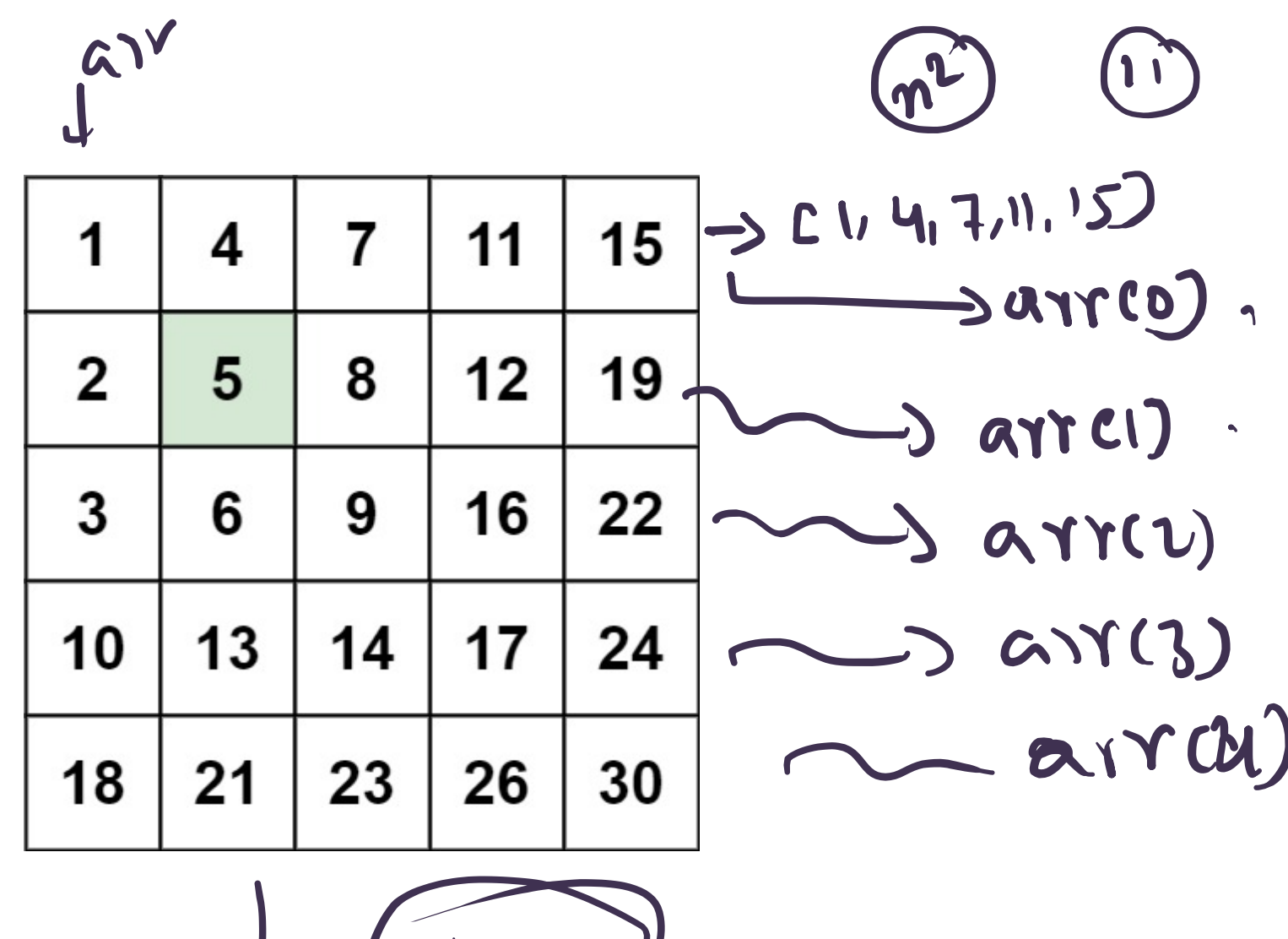
for (col=0; col<arr.col.len; col++)
if (col % 2 == 0)
for (row=0; row<arr.len; row++)
5x50 arr[row][col]

```
public static void main(String[] args) {  
    int[][] arr = { { 1, 2, 3, 4 },  
                    { 5, 6, 7, 8 },  
                    { 10, 11, 12, 13 },  
                    { 14, 15, 16, 17 } };  
    transpose(arr);  
    for (int i = 0; i < arr.length; i++) {  
        for (int j = 0; j < arr[0].length; j++) {  
            System.out.print(arr[i][j] + " ");  
        }  
        System.out.println();  
    }  
    public static void Transpose(int [][]arr) {  
    }  
}
```

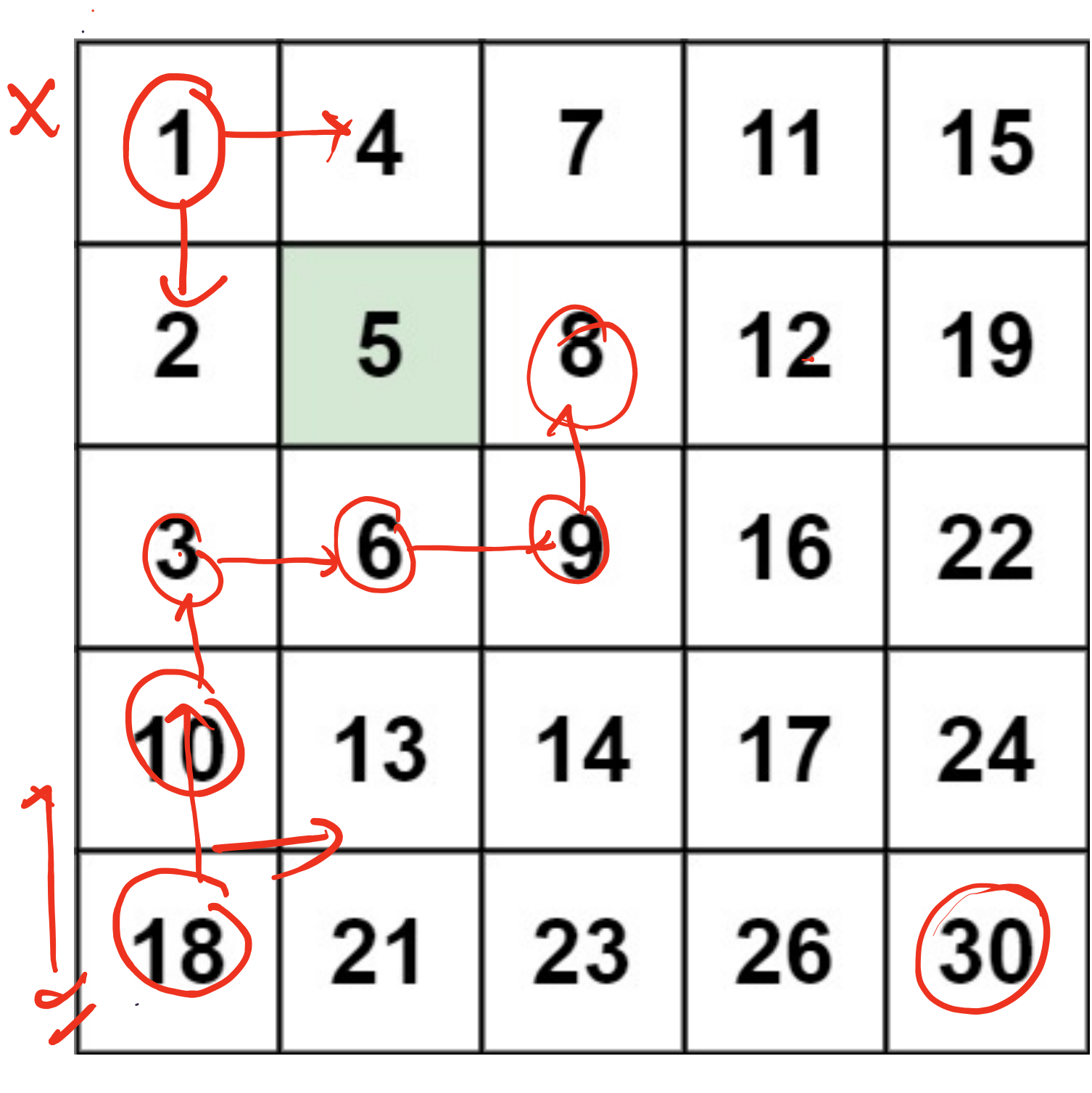


(0-1) → C1-0
(0-2) → C2-0
(0-3) → C3-0
(1-2) → C2-1
1-3 → C3-1
(2-3) → C3-2

for (i=0; i<arr.len; i++)
for (j=i+1; j<arr.col.len; j++)
swap (i, j) → C[i][j]



N log N
for (i=0; i<arr.len; i++)
if (BinarySearch(arr[i], target)) {
 return
}



Static case item=14
row=0, col=arr.col-1
while (row<arr.len && col>=0)
if (arr[row][col] == item)
return
else if (arr[row][col] > item)
col--
else
row++